

Sepideh Maleki

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Work Authorization: U.S. Permanent Resident (Green Card)

Research Interests

My research focuses on building the **virtual cell** — computational models that simulate how drugs affect cell lines at the molecular level. I develop machine learning methods to predict cellular responses to drug perturbations, with the goal of designing drugs for specific perturbation profiles. My work leverages foundation models, graph theory, and machine learning to advance biological and drug discovery applications, ultimately bringing more effective treatments to patients faster.

Selected Projects

Multi-modal foundation models for predicting molecular perturbation

I have been working on fine-tuning single-cell foundation models (FMs), pre-trained on tens of millions of single cells, to address molecular perturbation prediction. I introduced a drug-conditional adapter that allows efficient fine-tuning by training less than 1% of the original foundation model, thus enabling molecular conditioning while preserving the rich biological representation learned during pre-training. The proposed strategy allows not only the prediction of cellular responses to novel drugs, but also the zero-shot generalization to unseen cell types.

Developing a contrastive approach to explain (hyper)graph neural networks

I developed grXAI, a novel framework for graph representation explainability. grXAI generalizes existing score-based graph explainers to identify the subgraph most responsible for constructing the latent representation of the input graph. This framework can be easily and efficiently implemented as a wrapper around existing methods, enabling the explanation of graph representations through connected subgraphs, which are more human-intelligible.

Representation Learning on Hypergraphs

I developed a hierarchical hypergraph embedding system that can generate an embedding of a hypergraph with millions of nodes and hyperedges in just a few minutes while current approaches take days. These embeddings can be used in classification and prediction tasks on hypergraphs.

Classification and Prediction on hyperedges using transfer learning

I introduced the novel task of hyperedge classification. Using hyperedge classification as a pretext task with labels from a hierarchical community detection algorithm, we can perform hyperedge prediction. This self-supervised approach improves the state of the art in hyperedge prediction by a significant margin on standard benchmarks.

Education

PhD in Computer Science, University of Texas at Austin, August 2022

Master of Science in Computer Science, Texas State University, May 2016

Bachelor of Science in Computer Science, Kharazmi University, Iran, June 2011

Work Experience

Postdoctoral Fellow — Genentech (2022 - Present)

Mentored by Prof. Sara Mostafavi. Research on virtual cell modeling and molecular perturbation prediction.

Graduate Research Assistant — University of Texas at Austin (2016 - 2022)

Advised by Professor Keshav Pingali

Biomedical Data Science — Stanford University (Summer 2020)

Advised by Professor Dennis P. Wall

Software Engineering Intern — Google, Inc. (Summer 2017)

Developed a benchmark representative of the kind of code that Google executes.

Researcher — Texas State University (Summer 2016)

Graduate Research Assistant — Texas State University (2015, Spring 2016)

Advised by Professor Martin Burtcher

Graduate Instructor Assistant — Texas State University (Fall 2014)

Digital Logic, Foundations of Computer Science

Hardware/Software Tester — Sanjesh Afzar Asia (2012 - 2013)

Hardware/software testing of handheld devices capable of reading digital metering

Qualifications

Programming Languages: Python, C/C++, Java, CUDA, C#.NET, R, Matlab

Libraries & Frameworks: PyTorch, TensorFlow, LangGraph, LangChain, WandB, Scikit-Learn, NumPy, Pandas

Machine Learning & AI: Foundation Models, LLMs, Single-Cell Models, GNNs, RAG, Agentic Systems, Explainability, (Un)Supervised Learning

Honors, Awards, and Activities

ICLR'25 spotlight — Efficient Fine-Tuning of Single-Cell Foundation Models for Molecular Perturbation Prediction (MLGENX)

ICLR'24 workshop organizer — 1st Machine Learning for Genomics Exploration Workshop (MLGenX) at ICLR 2024

Graduate School Fellowship — University of Texas at Austin (2022)

AMS Mathematics Research on Sparse (Hyper)Networks — funded participant (2022)

Spring School — Lake Como School of Advanced Studies; Complex Networks (2022)

PSB session organizer — Graph Representations and Algorithms in Biomedicine (2022)

PPOPP artifact evaluation committee (2021)

Best Workshop Paper — WOSET (2019)

First Place — Conference of Southern Graduate Schools (CSGS) competition (2017)

Outstanding Master's Student Award — Texas State University (2016)

Computer Science Endowed Scholarship — Texas State University (2016)

Graduate Research Excellence Award — Texas State University (2015)